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24 July 1976
SUPERSEDING
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MILITARY SPECIFICATION SHEET

ELECTRON TUBE, BACKWARD WAVE OSCILLATOR

TYPE 7661

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Type "O", tunable 5320 to 5950 megahertz (MHz), 100 milliwatts (mW), minimum, air-cooled, integral magnet unipotential cathode.

ABSOLUTE RATINGS: (Note 1)

Parameter:	Ef	Ec	Eb1	Eb2	Ib1	Ib2	Delay Line Temp	tk
Unit:	V	V	V	V	mA	mA	°C	sec
Maximum:	7.0	-100	200	800	12	50	125	---
Minimum:	5.7	0	---	450	--	--	---	60

PHYSICAL CHARACTERISTICS:

Dimensions:	See outline drawing
Weight:	9.5 pounds maximum
Mounting Position:	Any

TEST CONDITIONS: (Note 2) The conditions below apply for all tests, except as modified in the conditions column.

Parameter:	Ef	tk	Ec	Eb1	Frequency	VSWR
Unit:	V	sec	Vdc	Vdc	GHz	Ratio
	6.3	60 min.	0	Note 6	F1, F2, F3 Note 7	1.20 Max.

REQUIREMENT OR TEST:

First Article Test - Required (See Note 8)

7661

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	First Article Test	See Note 9				
1042	Shock	15G; t = 11 ms	---	---	---	---
1011	Humidity		---	---	---	---
1031	Variable frequency vibration	F = 10 to 55 Hz; t = 180 sec; displacement .08 inch; planes X1, X2, and Y1	ΔF ΔPo	--- ---	0.3 10	MHz mW
----	Tuning linearity (1)	F2; See Note 3 (abcd)	Deviation	-10	-50	MHz
----	Tuning linearity (2)	F1 to F3 See Note 3 (abce)	Deviation	0	14	MHz
	Quality conformance inspection, part 1					
4250	Power levels	F1 to F3	Po	100	---	mW
4214	Cathode emission	Ef = 6.3 to 5.7 V; t = 120 seconds	ΔIk/Ik	---	15	%
1256	Grid current	Ec = 50; F3	Ic	0	15	mAdc
1261	Grid voltage	At power cut-off	Ec	---	-20	Vdc
1261	Delay line voltage	F1 F3	Eb2 Eb2	470 690	550 780	Vdc Vdc
4251	Power output variation (1)	F1 to F3	Ratio	---	3	db
1366	Inter-element resistance	Non-operating	R	1.0	---	Meg
1256	Anode current	F1 to F3	Ib1	0	10	mAdc
1256	Delay line current	F3	Ib1	---	45	mAdc
1366	Heater-cathode leakage	Ehk = +90 V; Ef = 6.3 V (No other voltages applied)	Ihk	---	5.0	mAdc
4251	Power output variation (2)	F1 to F3; over any 100 MHz	Ratio	---	1.5	db
	Quality conformance inspection, part 2					
----	Heater modulation	Ef = 6.3 V @ 60 Hz	ΔF	---	1.0	MHz
4027	Temperature coefficient		ΔF/ΔT	---	75	KHz/°C
----	Anode to delay line tuning sensitivity ratio	See Note 4	$\frac{\Delta F/\Delta Eb1}{\Delta F/\Delta Eb2}$	---	1.0	---
4243	Spurious signal output (1)	F1 to F3; excluding frequencies within ±2 MHz of the carrier; See Note 5	Carrier to Spurious	-75	---	db
4243	Spurious signal output (2)	F1 to F3 within ±2 MHz of carrier frequency	Carrier to Spurious	-45	---	db

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued					
4274	Voltage tuning continuity	F1 to F3	---	---	---	---
4266	Inter-Electrode capacitance	Between any electrode and all others except cathode	C	---	30	pF
		Between cathode and all other electrodes	C	---	45	pF
4030	Thermal frequency drift (1)	TA = -30°C ±2°C; F3	ΔF	---	1.0	MHz
4030	Thermal frequency drift (2)	TA = Room temperature; F3	ΔF	---	0.5	MHz
D-30(b)	Dimensions	See Figure 1	---	---	---	---
1301	Heater current	Ef = 6.3 V; tk = 180 ±15 seconds	If	0.90	1.10	A
4243	Signal noise to ratio		S/N Ratio	90	---	db
	Quality conformance inspection, part 3					
	Intermittent life test operation	Group D	t	1000	---	hrs
----	Life test cycle	Preheat: 60 seconds Radiate: 7½ hours Off time: 29 minutes Change frequency 100 MHz 24 hours; starting at F3	---	---	---	---
	Life test end point:					
4250	Power levels	F1 to F3	Po	80	---	mW

NOTES:

1. The absolute ratings must not be approached or exceeded if the specified life is to be obtained. The absolute ratings are limiting values beyond which the serviceability of any individual tube may be impaired.

2. The voltages must be applied in the following order if sequential or delayed application of voltages is required.

    1. Heater
    2. Grid (may be eliminated from sequence if zero)
    3. Delay line
    4. Anode

    The grid, delay line, and anode voltages may be applied simultaneously and all voltages may be removed simultaneously with no detrimental effect. Existence of anode voltage in the absence of delay line voltage can result in permanent damage to the tube.

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NOTES: -Continued

3. Tuning linearity shall be measured as follows:

   a. Plot the actual tuning characteristics (MHz vs Volts).

   b. Draw a straight line between the 5320 MHz and 5950 MHz points.

   c. A smooth line drawn through the center of the fine grain variation of the tuning characteristic shall be referred to as the smooth curve.

   d. The distance from 5635 MHz on the smooth curve to the frequency on the straight line for the same voltage shall not exceed the maximum value specified.

   e. The distance from the measured tuning characteristics to the smooth tuning curve shall not exceed the limits specified.

4. The anode voltage sensitivity shall be less than the delay line voltage for all points in the hand. The test shall be performed by varying the anode voltage from minus 10 volts to plus 10 volts of the value stamped on the tube and measuring the change in delay line voltage required to obtain the same frequency at both anode voltage levels.

5. At carrier frequencies from F1 to F3, the ratio of spurious to carrier power shall be measured between 5120 MHz and 6150 MHz.

6. The anode voltage for this test shall be adjusted to the value indicated on each tube. This value shall be between 70 and 150 volts.

7. F_1 = 5.32 GHz
   F_2 = 5.62 GHz
   F_3 = 5.95 GHz

8. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this specification sheet. In addition, first article sample approval shall include satisfactory demonstration of tube-to-system compatibility. Invitation for bids should provide that the preparing activity reserves right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

9. These tests need not be performed except when First Article testing is required by the contract or order.

Custodian:
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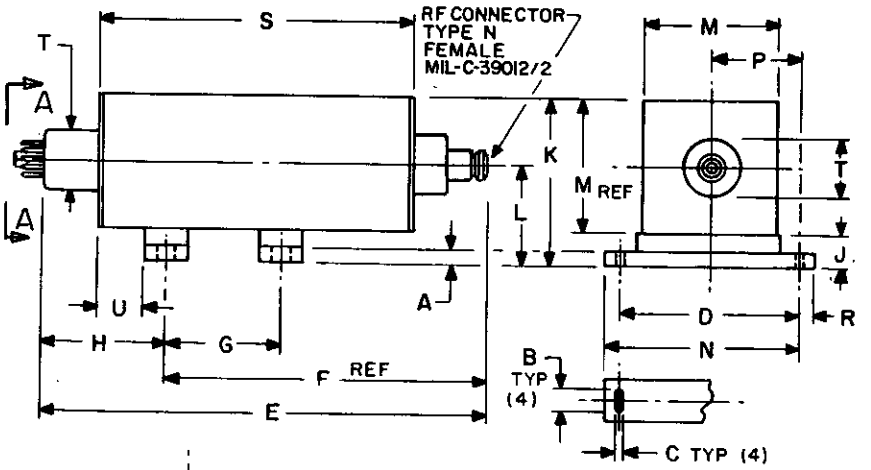
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Air Force - 85

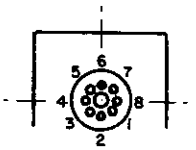
Review activity:
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User activity:
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VIEW A-A
OCTAL BASE CONNECTOR
(ORIENTATION AS SHOWN WITHIN ± 20°)

OCTAL BASE CONNECTOR			
PIN	CONNECTIONS	PIN	CONNECTIONS
1	DELAY LINE & COLLECTOR	5	CATHODE
2	NO CONNECTION	6	HEATER
3	HEATER	7	ANODE
4	GRID	8	NO CONNECTION

DIMENSIONS

SYM	INCHES			MILLIMETERS			SYM	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX		MIN	NOM	MAX	MIN	NOM	MAX
A	.365	.375	.385	9.27	9.53	9.78	K	3.625	3.750	3.875	92.08	95.25	98.43
B	.375	—	.400	9.53	—	10.16	L	2.150	2.250	2.350	54.61	57.15	59.69
C	.180	—	.195	4.57	—	4.95	M	2.950	3.000	3.050	74.93	76.20	77.47
D	3.950	4.000	4.025	100.33	101.60	102.24	N	—	—	4.312	—	—	109.52
E	—	—	9.938	—	—	252.43	P	1.900	2.000	2.100	48.26	50.80	53.34
F	—	—	7.188	—	—	182.58	R	—	—	.312	—	—	7.92
G	2.540	—	2.590	64.52	—	65.79	S	—	—	6.995	—	—	177.67
H	3.125	—	3.250	79.38	—	82.55	T	—	1.375	1.400	—	34.93	35.56
J	.725	.750	.775	18.42	19.05	19.69	U	1.000	—	—	25.40	—	—

NOTES:

1. Finish: Black anodize, satin finish. Per MIL-A-8625.
2. Maintain 4 inches clearance from magnetic materials. Do not subject to magnetic fields in excess of 100 gauss.
3. Delay line and collector are insulated from the external case.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1 Outline drawing for tube type 7661

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